

ActInf Livestream #012.0: 2020 Review and 2021 Directions

Livestream | 1:33:41

hello everyone

this is the active inference live stream

i'm daniel friedman today it is december

30th

2020. it is active live

active influence live stream 12.0 and

this is uh

hopefully going to be a recap on 2020

and a look to 2021

it is a doubly auspicious episode number

because the digits of 012.0

rearrange to 2021 and also the clock is

striking midnight

so there's no paper 12. we'll start with

paper 13

next year welcome to

the active inference lab everyone

we are an experiment in online team

communication learning

and practice related to active inference

you can find us

at our website twitter email

youtube or keybase team and username

this is a recorded and an archived live

stream so please provide us with

feedback so that we can

improve on our work you can say

something in the live chat

or you can leave a comment

all backgrounds and perspectives are

welcome here

and as far as video etiquette for live

streams mute if there's noise in the

background

raise your hands so we can hear it from

everyone and we'll use respectful speech

behavior

today in active inference live stream

12.0

the goal is really to close out 2020 and

look to

2021 for all that that means

of course just like all the dot zeros

it's a context and an introduction

to some ideas it's not a review or a

final word

the punchline is that 2020 hindsight

looking back on it

is 2020 foresight and that's all we have

so let's figure out how to make the best

of our 2020 hindsight

so that we can have 2020 foresight

here are the sections of today's video

first we're going to talk about what the

active inference lab

is becoming in terms of what did happen

for each of the three projects in 2020

and what will happen for each of the

three projects in 2021

then we'll go through each of the active

inference live streams a single slide

per number 1 through 11

and just recap one figure from each of

the papers because it's a good way to

just

look at all the different topics we've

heard about

and remember how many interesting

perspectives and especially beginners

perspectives we heard

along the way because we're all learning

here and we all learn so much this year

then i'll raise up a few great points of

feedback that were raised by the
community
and we always welcome people to leave
comments and suggestions and if they
want to be anonymous they can
let that be known or otherwise we're
just really
looking to rise up what people are
saying and wanting to learn
and then let's just figure out how we
can make those things happen and they're
both
really good suggestions we'll talk about
then
uh there'll be more uh just sort of a
grab bag at the end with some
perspectives on 2020 and 21 moving
forward
so in the upcoming year there will be a
lot to expect and work on together
be sure to get in touch if you want to
participate or get involved
so for today's roadmap instead of the
sections of a paper because there are
no sections for this paper we're going
to be
inferring back on 2020 and we're going
to be
actively moving into 2021. so whenever
you're listening to us certainly it's
either at the same time or later than me
and that means that you're in 2021 or
later or about to be rolling into it
so how can we look to the past the
present
and the future how can we understand
where we've been to understand where
we're going

in this example with a background
picture of the rear view mirror of a car
it's like you're driving the car forward
but you need to know what's behind you
you need to know what's around you
so all of these things are important
spatially temporally

starting with the part about the active
inference lab

uh this was really something awesome
that emerged during 2020

with the special help and collaboration
and really was a true colony effort by
alex yvonne sasha rj and just

several other people who really made it
happen

this year with active inference so big
appreciation to them

and with the more people we've had
participating

and increased focus on formal structures
as well as values of transparency

accessibility inclusion etc we

resulted in the active inference lab so
not something that any of us thought we
would be

involved in at the beginning of 2020 but
again a year that changed so many things
for so many

people so if you go to
activeinference.org you'll see

this page which contains information on
the bottom about how to contact

our group through our shared markov
blanket

of these identities and so these are
going to be

always like organizationally held

identities online then you'll also see
at least for the time being this call
for collaboration for 2021 so that's a
form that we put out an open call for
collaboration in
in three areas that we're about to be
talking about which are in the domains
of education
communication and tooling and uh
so far on that call we've had a good
number of rcps so we're really
happy that people are excited about it
and we're
open to not just participation but input
on how to
organize and make it in line with how we
want to see a decentralized research
community be and how to
how to have a good active inference
community
so the three areas of project that we're
gonna recap
2020 on and then look to 2021 on
our education communication and tools
and this is just
from the copy uh screenshot of the
letter
that is the url on the right side which
is the one that's on the website so
that's where to see this information
okay
the first project is education and it
was said as our cornerstone project for
2021
is a community developed open source
active inference book of knowledge or an
aibok
and educational courses as a result this

project will entail ontological so
ontologies related to structuring
knowledge
methodological that's processes and ways
of learning and doing research and
methods for various
areas and pedagogical related teaching
research and development
let's go a little bit into that because
i actually did not know about the book
of knowledge approach before i met alex
and yvonne
and their systems engineering
perspective
this is a graphic that alex provided of
the
systems engineering book of knowledge or
cbook
s-e-b-o-k and the scope of
this area and uh suffice to say that the
systems engineering
people have a really unique approach
that fuses the
multi-scale approach that
helps us deal with complex systems with
an engineering focus
on defining interfaces and standards
and they took a very standards driven
yet flexible approach
as far as i can tell though i'm less
familiar than they are
in structuring their educational
material and so we can see
kind of around the edge we have various
things that are peripheral skills i
believe
related to the core competencies so for
example the left side people

competency knowledge so that's critical
and even though
you're going to be in this area training
somebody in systems engineering
it's also uh critical that they have
people competency so not just
social workers need that skill everyone
needs these skills
operational environment knowledge on the
right side that could almost pertain to
anything
and then in the middle there's these
core domains of
the curriculum and this curriculum
facilitates
easy onboarding easy referencing people
can point to a specific document
a specific web link and just say in this
section this was what
the way it was said was and that
includes
uh the foundations the introduction
foundations
the applications management
enabling factors related disciplines and
implementations
so perhaps you can see where we're going
with active inference here is
sep okay in their environment and this
is the part
that i think is built out to an extent
that's broader than merely just a
curriculum so the previous
slide this one contains information
about the way that they structured their
curriculum
as far as i see it but i have not taken
these courses in the way that alex and

yvonne have

where it gets really interesting and

unique

as far as a transferable structure is

this

scbok the environment plot so here we

have the community

which includes multiple types of

stakeholders it includes academics as

well as non-academics that just like

many other research communities do

within the community there are

professional societies professional

societies

have structured relationships using

ontologies

with specific types of other things such

as certification programs

competency models workforce development

initiatives so getting people involved

we could also imagine a participation or

a just

public outreach or public communication

initiative so there's all kinds of

initiatives

that can be related with an

organizational ontology

to our community and then we can also

think about even

products maybe there's an open source

robotics uh or an open source simulation

framework

that our groups develop so we're going

to have formal relationships within the

environment

the scbok is part of a developed

ecosystem

not just of a curriculum that they're

offering but also certifications and professional societies etc could active inference have a community that is participatory so focused on people being active it's also professional even though we recognize there's people from wide variety of career stages we want to keep it professional we want to be trans-disciplinary which means diving into the specifics where we need them but not worrying about what the delineation is between disciplines and also have an aspect of modern organization like this clearly has as well as recognizing the global and cross-sector practices that are going to come to intersect in active inference already it's seen as a meeting of two different perspectives when subfields of one area come to meet and in this domain we're talking about philosophy and we're talking about mathematics and so sometimes there's uh some fundamental issues that rise up so how do we deal with these truly fundamental issues that we often bump into as individuals or groups well we all we also can uh get around those or at least structure around them to structure space to hold those debates within within a more built out ecosystem like this and that can

make us have healthy norms and visions
for our community
so of course join us to help because it
has to be enacted it has to exist
so join if you're interested second
project
we will continue the weekly active
inference live stream this being example
of
as an accessible and participatory space
for learning about
active inference we are also developing
new formats here such as seminar
presentation dialogue working group
and short video so here's the
communications
project so in 2020 we had
on our youtube channel we had 27 or so
participatory live streams there were
dozens of really great participants
so that was very special
uh over the course of the year we made
many
uh improvements in the technicalities
of the project so early in the
project at the very beginning we were
just using jitsi and then live streaming
directly from jitsi
which is a variable strength connection
and
now i'm utilizing obs software so
figured out what kind of parameters work
there and hopefully that's resulted in a
better
experience but beyond the technical
purely part
there's also been an emerging practice
for how can we engage in accessible live

streams and we don't have the answer
and we think that there's so much more
space to improve on
and want to hear people's feedback for
but we develop things like
checklists for participants so that they
can run through and understand
uh hopefully how to be prepared for
being on a live stream and that's
facilitated
hopefully some discussions that
otherwise couldn't have happened without
that type of
gentle normie and at this link
you can learn more about our 2021
plants so for 2021 for the active
inference live stream we're gonna be
doing it every single tuesday from seven
to nine
a.m pacific time whatever it is pst for
now but that will change so pacific time
and for the uh or maybe we'll see about
that
i guess we'll think about it uh for the
schedule
and participation instructions just go
to this link
and of course just let us know if you
want to participate
in any way in 2021 through this
means or any other or if you want to
have a special event or something that
we could do
on this stream channel if you go to that
link on the left side you'll see
this spreadsheet here's the current
state of it up till the middle of april
where we have the authors of the papers

will all be on at least

once and on the top are the steps you
need to take

if you want to participate as well as on
the top right

two awesome resources by jared and
barron

that are like hubs or repositories of
information about active inference so if
anyone has suggestions they can also
suggest something that should go up
there on the top right so we want this
to be

a good landing page for people to just
go to this short link

comes to the spreadsheet here's the
curriculum here's the year if you want
three months out you want to read a
paper that's interesting

let us know if you want to suggest a
paper we can also

do that here's the last area
of the active inference lab for 2021 and
2020

is the third project which is tools
so this says following our work on
active inference in online teams
we will continue development on tooling
such as knowledge management systems
collaborative research models
simulation of digital workspaces and
onboarding tools

here's a figure from the paper that
team com co-authored so
alex yvonne sasha rj and myself
and this figure is showing the online
team

in sort of a flowchart layout

that's reflecting the specific kinds of
actions that connect specific kinds of
actors
and it's basically uncontentious
that the team members are communicating
with a certain database or with various
databases and through these internet
backbone intermediaries they're
communicating or having specific types
of defined communication
and that can also communicate to
external databases
and our uh question that was guiding us
was could we think about those
computational actions
as basically going through a markov
blanket
slash influencing things across markov
blankets however we want to think about
that
passing through shared niches of
information there's a lot of ways
that the active inference framework to
us made us wonder about the online
context specifically
and uh this paper which explored a few
other ideas about the kinds of
technologies that would enable active
inference for online teams instead of
just
the kinds of tools that we have now also
this was just one of the other
figures that show the shared world
are informational niche that's like the
chat room that both people
are in that's their shared niche and
then they can send and receive from that
chat room

and that plays into their model it
updates their priors
on some aspects of their generative
model that leads to policy changes which
reflected in changes in action
here's where we want to go from that
paper in september
to something that could be more of a
minimum
viable usable for 2021.
uh as we head there and as we build
this uh tooling out
we want to first and foremost keep the
individual
in mind the human as the user
and we currently have a bunch of people
who've RSVP'd to work on this project
three
so i don't have any of the details on
that we're going to
basically co create that during january
and beyond but here's just some of the
questions and then
other scales of analysis that we can
build upon
hopefully so they're more just like
open-ended
things that maybe the team three will be
keeping in mind and i hope that we also
add to this list of questions
so one question is about how digital
user interfaces can shape and design
regimes of attention so we talked about
cultural regimes of attention
individual regimes of attention from the
ocular motor to
the uh rumination the strange attractors
of thought

and are we going to have the basins of
our attraction
drawn to the news feed or are the basins
of our attraction going to be
drawn towards deep work that's a
question and
are we going to design the niche to
facilitate individuals to choose
how can tools be useful and intuitive
for all kinds of people
and purposes and that's really important
because people use the internet in so
many
different ways and there's so many kinds
of people using the internet so
especially
if we want to be developing powerful
tools we want to make sure
that they're meeting people where they
are
another level of analysis to start
broadening beyond the
individual person is at the relationship
and the team level
so there's a lot of questions about how
could
online teams and projects be innovative
productive and resilient
in ways that were defined potentially
within the active inference framework
for example controlled
novelty search or the ability to
discover
innovative solutions on rough landscapes
or on
landscapes with very infrequent rewards
another question is about even just
given the current software operating

environment today
how could active inference on the back
end using just purely
analytical frameworks lead to useful
tractable
and meaningful analytics for teams for
example have an underlying generative
model
of teammate states instead of just doing
descriptive statistics and saying
here's the distribution of how many
messages these people sent
you could use some other type of latent
state modeling
to think about those emissions from
those peoples as
actors and uh as agents whatever you
want to
draw the mark on a blanket around you
could even do that for
individuals for which you don't see all
of their
emitted states for and that can be
helpful in team context
and then the last level would be at the
communities and the ecosystems of
communities of practice level
a big question is how can online
communities be welcoming accessible
inclusive and trusting we want to have
high trust online settings
and hopefully have the hard like
cryptographic and soft like people
interfaces that allow that to thrive
and how can we have healthy norms for
facilitation
mental health communication in our
active inference

community but also in other communities
and
share the emerging practices that work
across communities while also
understanding that just like different
things are gonna work for different
people
different norms are going to work for
different communities so it's not
that one approach has to catch on with
anyone except for the people who like it
and always coming back to the question
of how
active inference might be able to inform
the way that we design these multi-scale
digital ecosystems
okay so those were the three projects
and if they
sound interesting you should sign up
and if it's past 2021 then
check back and there'll be another form
now we're going to go
one figure per 1 through 11
of the active inference live streams
kind of just to recap where we've been
to remember the kinds of debates
that we raised the the tensions and the
integrations that occurred throughout
this year just to juxtapose it get some
of our
laden connections uh reactivated
the first active inference live stream
paper
was narrative as active inference the an
integrative account
of the functions of narratives and this
stemmed
really from our initial curiosity as a

group

about how narratives might be understood

from an active inference context

because we all understood less than

about the formalisms underlying it to

wherever we were at we were at an

earlier point in our understanding

and the part that was drawing us in as a

team

was about teams and narratives and about

how narratives for teams

relate to how they commute it's caught a

communicate so cause and effect with a

narrative in the team

and the figure that captured it for us

was that in the online case the

epistemic resources and salience queues

are formally defined so if you're

talking about two people who have an

edge via a text messaging thread it's

formally defined you know a hundred

percent of when each message was sent

maybe you can read every message if

you're one of the two people

hopefully no one else or if you're

someone else

maybe you can even at least see the

patterns of their communication

the timing of them or the timing and the

length or some other attribute

but the point is that in a digital

system it's formally defined so that you

can completely capture

it's like a formal state machine in the

niche in real

world settings it's very difficult to

capture

the nuances of the situation in a way

that doesn't mean you can't classify behavior with a 99 or 100 accuracy it just means that there's uh still aspects of the system that are just empirically reduced in dimensionality because you're making measurements on a real world system whereas the internet system it's like it's its own digital twin it's already the digital thing so it's way easier to formally model and what interested us was just the way that individuals who could have shared or divergent narratives who even knows how to calculate that how do you calculate narratives and measure them they would be involved in a sense action active inference loop as would their colleagues and teammates and then their teammates that could be anonymous commenters in a forum or a comment thread it could be your professional colleague who is on a active inference lab project with you and these shared resources how could those be in cause and effect and how could this interface be designed so that it shapes the kinds of regimes of attention and deliverables under the right timeline under the right budget in line with values helping everybody these are the kinds of things that could be designed through the regimes of attention which are none other

than the focus of the team
the second paper was is the free energy
principle a formal theory of semantics
meaning from variational density
dynamics to neural and phenotypic
representations
here maxwell uh really
helped us understand as always
some of the finer points philosophy
intersecting with math of this
paper and this is a summary
figure from the end part one and it
relates to
the connections between different ideas
that come into play or fall under the
umbrella or scope
of free energy principle slash active
inference broadly
and the key node is on the left side
this computational dynamics
so it's not quite computationalism like
everything is a computer uh it doesn't
have to have that kind of a
computationalist overtone
it's just a realist or pragmatic
understanding
that we are going to use computers to
model the dynamics which are the changes
through time
of these things we could also have like
an experiential dynamics a
psychodynamics
to be sure but any kind of computational
model is computational dynamics so this
is like saying we're going to be able to
phrase it with math
and computer programs so not too much of
a

pill to swallow there but it uh starts
getting interesting
with uh the integrations of some
classical and non-classical attributes
of computation
so starting on the top mathematical
functionalities are implicit
in computational dynamics computers are
based upon equations
equations and computers are deeply
related and the way that that bears upon
the specific uh framework that they're
talking about here which is the
variational density dynamics
is that there's a mathematical function
which describes the free energy
functional
that's a function relative to a
generative model
of a few different things things that
we're going to be describing soon
the second node on the middle is the
specific algorithms
again shouldn't be surprising algorithms
are computational in nature
the way that that gets implemented in
active inference is via
this gradient descent on the variational
free energy function
functional and then that goes back to
the mathematical function in the first
node
okay now the middle node in the middle
representational
structure ontology that's the structure
of how things are in the world
so then that plays out in active
inference with this idea of internal

states

so let's just say that my internal state is the estimate of whether it's sunny or not sunny outside so there's many ways that the sky can be there's many shades of moonlight and sunlight and it could have many many degrees of statistical measurement but the internal state for me as somebody who cares only about whether it's sunny or not is a two-state categorical model so organisms embody the statistical regularities of their niche but they don't need to have every single degree of freedom the whole point is that they are a coarse graining or dimensionality reduction of that niche otherwise they would be this one to one that's the map that covers itself so the internal states are a representational structure we're going to get into representationalism in a later paper but they are a representational structure at the very least in the statistician or in the investigator's analysis of the niche okay fourth note computational processes seems that that's under the computational dynamics and that's where active inference comes into play which is the computation of this active inference uh loop as sort of a structural computation that's played out on bayesian nets or message passing

or other kinds of graphs these forney
factor graphs
things that would be great to have
colleagues who
know more mathematics come onto the
stream let's have a
special presentation or discussion on
some of those mathematical
or computational formalisms the final
one
and this is where i think an entry point
lies for many
is this ecological component now that's
formalized on the bottom right
under this model as the dual information
geometry
which is this interface that's defined
by the organism
at its boundary is some type of
blanket some type of isolation there's
various ways i think we can think about
it
and it's clear that the markov blanket
concept
people are discussing and generalizing
and critiquing every day so
let's figure that out another day i say
suffice to think that for some systems
especially locally
it's the interface that defines uh the
inner
uh interactions between these two sides
so formally that's reflected as dual
information geometry
but qualitatively that's where the
inactivism and so many other parts of
the relational
part of active inference come into play

in paper 3 a world unto itself human
communication as active inference
jared joined i was very memorable and
here was a figure
that kind of like many other things
we returned to similar figures again and
again
but this one brought us back to this
core
loop internal states and external states
then sense and action these are the four
states
now here's the blanket goes down the
middle that's why
the metaphor of a quote blanket uh like
a fabric blanket or i don't know
what kind of blanket
whether it's that as a metaphor or the
tapestry or
whether you just want to think about it
purely without any words just in terms
of the exact
computational influences in the model
that's where
the debate in lies and then the symmetry
that was described on the previous slide
the dual symmetry is that the internal
and the external
are ultimately just following the same
computational structure it's sort of
like how
you could have experimental group a and
experimental group b
but you could have called them the other
letter that doesn't mean you could have
switched which one got the drug and
which one got the placebo
but you could have called it the other

letter so there's a symmetry
with the labeling though there's not a
symmetry with the treatment
and so this four
state model is just something to be
returned to again
and again what are the states what are
the connections between the states
and then leave space for the unknown is
there something that transcends this
ask if there's another connection if
there is
it can be added in but if there isn't
this is a parsimony's descriptor and it
allows us to do this fractal nesting and
a few other things
another part of this figure was
going one layer into depth onto
how each of these states were defined
and how each conveyed
specific aspects of what the system was
doing
in terms of a drawing on various
frameworks using terms from
bayesian statistics other times using uh
terms from information theory and uh
sensation we can just map them really
quickly is the
uh sensation is is uh trying to minimize
the agent's bound on surprise
sensations play into internal states so
here's
the uh photons the eye is seeking the
uh is trying to bounce surprise on the
visual world
the internal states are optimizing the
agents
bound on surprise which is basically

saying given
the input the upward uh flowing input
how can we tighten our expectations
through
state updating and optimal policy
that leads to agent action to minimize
the agent's bound on surprise
so that's why sensation and action both
have the same
goal of bounding surprise that's the
theme
bounding surprise as defined by what
surprise is from this
internal state model perspective
and then the niche is also optimizing
the niches bound on surprise
now i think that uh
this symmetry is the dual information
geometry
but it remains to be seen a little bit
how that
would play out in a real niche i think
it'd be an awesome modeling
project or a discussion about how we
could think about what it means that the
niche is bounding its
surprise does that mean it returns to
attracting
sets of nitrogen concentration
so it's almost like if the biological
community
goes to a certain concentration in a
certain community variable
then there's a drawback what does that
exactly mean
there's a lot of these kind of gaia
related papers maybe that
could be related but this paper was

about communication
and so we went from the narrative sort
of down a level but also up a level
thinking about the turn taking
and about the way that communication
encodes
meaning and conveys meaning and about
culture and about the way that culture
structures
the meaning's interpretation the fourth
paper
brought us more directly to culture
itself
and that was the paper cultural
affordances scaffolding local worlds
through shared
intentionality and regimes of attention
so that's when we got into the regimes
of attention
mindset that was on our regime of
attention
and this was sort of one of the latter
figures of the paper
and it conveys this idea of regimes of
attention
quite well so it's sort of a schematic
for a generalized computational
process one that the authors posit
may be plausible given neural
architectures
in people and also potentially
implementable
amidst a family of other ways that it
could be implemented in
computer systems and this relates to
the uh bottom-up and top-down
processing and it also relates to
the bayesian brain multi-level bayesian

brain multi-level bayesian modeling
here's how it works the green nodes and
this isn't
maybe a colorblind uh visual but the
right
and green nodes are passing predictions
to computational units on this left side
of the column
the computational units on the left side
of the column pass their prediction
errors
up so this would be the the
bottom most level of this uh layout it
could be side to side you can use in and
out metaphors there's other ways to
think about it they're not always
like a totem pole it could it can be a
heteroarchy but in this one it's just
laid out like this
and it reflects how like the retinal
cells
let's think passes to the next link in
the chain
the predictions that uh
it's going to give to the next
computational unit
and then the errors the expectation
which is the deviation from the
prediction from the higher up node
is passed up so it's saying if it ends
up that this
is passing the exact same prediction of
the light past hitting it
and this one's getting the exact same
prediction then it doesn't pass any
differential
so that's predictive processing where
basically the downward

cascading prediction towards the sensory
cue
is kind of canceling out expectations
but residuals or errors propagate
upwards and upwards and upwards through
these abstraction hierarchies
to the highest level in this
computational framework again i don't
know if there are other or future
frameworks the topmost level
passes this yellow area arrow which is
the modulation of
attention so that attention arrow goes
from the highest
uh prediction not
uh yeah the highest most prediction in
error coupled unit
and passes it to this blue node which is
weighting the precision of different
kinds of
evidence different levels of the
hierarchy so let's just say we have the
visual
level and then we have a cultural level
the cultural level says all cars
are red and then
you see driving really fast a car and
maybe if you would have been uh not
in such a strong cultural context of
thinking that cars were red
you would have seen it as orange there's
all kinds of ways that our priors
can interact with and ultimately
structure the sensory cues that we
receive
and so this model captures that
modulation of attention to either a
cultural narrative let's say at a higher

level of abstraction
or a more sensory narrative
and that modulation that precision
weighting
which could be performed by changing
parameters in a computational model
or for example releasing a
neurotransmitter like dopamine into a
certain
area of the brain or type of synapse all
these neuromodulators
other kinds of non-neurotransmitter
neuromodulators
may over different time scales
differentially weight different levels
of this computational hierarchy
and that can be considered the regime of
attention
because the regime of attention may
hyper focus
on small things it may
due to precision weighting leading uh to
changes in state estimators or like the
relative
salience of different attributes or the
plausibility of different
features people's regimes of attentions
may be making them suffer
or maybe dysfunctional for a society so
different regimes of attention
may be things that people want to
pursue or not and that was what
this question of designing regimes of
attention was so
there has to be a regime of attention
what is it going to be
what shapes it the fifth paper we read
was multi-scale integration

beyond internalism and externalism
and this was a great paper
as really they all are and
it was about this philosophical debate
between
internalism and externalism which are
sort of like two
extremes on a continuum or
you can think of it probably different
ways just like any other philosophy
debate
but to put it simply the internalists
highlighted the role
of the internal processes that is
internal to the organism let's say
such as things that happened in the
brain and they highlight the importance
of those things
such things that happen in the brain
things that happen in the cell things
that are in the genome
these internal and usually smaller
things
versus the externalists who tend to
highlight the functional
uh roles of external things which tend
to be bigger things like
social groups or city structures
or civilizational level forces these
broader level things so it's a little
bit like the reductionism to holism
debate
but it's kind of framed in an internal
to external of a blanket
debate because everything is nested
there's not really any debate about that
it's a great observation to make but
really it's the starting point which is

why models like this are so exciting
because things are nested we know that
the only way that things wouldn't be
nested would be
if somehow there was a bigger thing
inside of a smaller thing
but that would be definitely
definitionally impossible
in the spatial world not in some other
dimension maybe
but in the spatial world it can't happen
so in
the kinds of systems that we want to
model uh if not many many more than that
it seems to hold this nesting so these
kinds of models
take that nesting seriously they say
look we're not going to be able
to simulate down to the lowest level
that's not even helpful
for example doing a protein simulation
of the vibrating atoms can only be
simulated for
a couple of seconds uh milliseconds
or of time and it's usually a very high
computational cost so it's not even
like you can go to every protein level
sometimes you can coarse grain we
we we can cheat every once in a while
okay so how do we
do that correctly and they use
this idea of multi-scale nested markov
blankets
to get at a way in which we could
perform multi-scale integration and by
doing so
integrate across internalist and
externalist perspectives on systems

function

and so the way that this works is that

there is a

markov blanket around hidden states and

this markov

blanket or any other kind of blanket or

metaphor you want to use

it's a hexagon here has incoming

variables

those are the sense states and outgoing

variables

those are the action states or

dependencies or relationships

and it can be coarse-grained no matter

how much complexity is

inside of here in the hidden states i

might want to have a two-state

model friendly or faux or i might want

to have a 50-state model

which uh you know percentile of this is

it with respect to this

or might want to have a 5000 state

that's a question about the mapping

between the observer and this system

because we're not saying that we're

knowing that system we're calling them

hidden states because

we want to be clear that we're choosing

which attributes to model

latently within this because all we can

observe

are which sense data are being projected

and

which action states are being emitted

now because of this coarse graining

it's possible to perform some

mathematics let's

hear from the authors and other

colleagues who know more on this
to think about that as a particle
within an ensemble at a larger level
of similar things so
here is an ant and it's getting inputs
chemo sensation through the antennae
detecting the chemicals
mechanoreceptors on its body with the
hairs detecting interactions
all these different things and then
internal states
that are in the ants reflected by the
way that it processes
these input streams of uh sensory cues
resulting in the updating of action
states where its body is
how its glands are secreting different
kinds of things
so that's at the single nest mate level
but the nest mate is
in a colony of other nest mates and then
depending on how evolution shapes it not
to get
too off onto another topic that's the
level where
one could expect to see rival risks
versus non-adversarial dynamics
is where and how selection is shaping
across this multi-scale system so this
paper was really
great and again maxwell just really um
good insights these are good
foundational papers and
i think they set the stage well for
talking about multi-scale
systems in paper six
we went a little bit deeper still into
this

two density idea this idea that uh
the paper was a tale of two densities
active inference is an active
inference and it relates to these two
densities that are being passed across
the blanket through the holograph
whatever it is
these two densities q and p uh action
and perception everything here that's
what embodies the relational insight
and inactivism and ecological psychology
everything
that fundamental relational perspective
is related to passing these two
functions
across an interface that's the simplest
way to think of an interface what goes
from side a to b
and vice versa that is what defines the
the minimal interface
just like computers with apis so
what are these two densities because
in what they actually are is the secret
of
how to move beyond internalism and
externalism
as per the previous paper and
this paper laid out a
flowchart for how variables are related
and implemented
from a structural mathematical
perspective
in a way that shows
us how because active inference is
inactive as suggested in the title
it relates the two densities action and
perception
so the claim is because active inference

is

enacted it builds upon the philosophical
framework of an activism

and therefore integrates sense and
perception action and sense

just like those inactivists ecological
psychologists

just like they have been what they have
been working

on with the sense action integration
this

mathematical structure captures
the way that those two densities here
across the interface are linked

so because active inference is enacted
and it includes action in the loop as
part of the causal structure of the
model

and part of the fundamental
way that this is laid out at a systems
level it builds upon

the inactive insights

so interesting paper and this

graph is something that we'll return to
in another uh later paper but

it related how free energy expected free
energy relates to

policy selection policies

have predicted influence on the b matrix
which is an estimate of how hidden
states of the world

change through time those states result
in observed

sensory outcomes through time and those
observed sensory outcomes through time
map onto the way that the other side of
the system

is going to change through time here's

its b
matrix as it changes on the other side
of the interface and
the energy minimization of that let's
just say niche
in this case it enacts its own policy
little pi and little pi
ends up influencing how the policy of
the top
agent influences its estimates of the
hidden states of the world updating
so that's the feedback and the claim is
that this
is an activism specified
formally or one way to specify it
formally
because it ties action and perception
together so
maybe it's simple maybe it's not but
it's an interesting thing
the seventh paper was variational
ecology
and the physics of sentient systems and
this was another
great paper so on the left even though
i believe there was a claim that maxwell
like walked it back
we can remember or listen to where he
said that but
just schematically or heuristically it's
just describing
how systems that exist over longer
periods of time
also can exist over larger spatial
scales
so this has to do with order emerging
from lower level
activity and there are things that

propagate

at the subcellular level like a

secondary messenger a calcium wave but

then

over larger and larger time scales and

spatial patterns

there is higher and higher orders of

function

higher and higher levels of reduction of

uncertainty

by understanding the underlying

generative model and then

on the bottom here are 10 bergens for

questions

maybe they could be laid out in that

arrangement it doesn't really matter

they're really more of a two by two grid

but we talked about that in seven and

it's a great framework

and it builds on the 2018 paper

answering short english question

by by ramstead at all which is really a

great one too

and then on the right side is sort of a

building upon

the multi-scale insight

from paper five and it shows these

particles

being linked in a way i'm not exactly

sure with the math let's

hear from the authors or anyone else but

what it looks like from the coloration

is that there's that same multi-scale

insight

but there is also the organization of

agents that may want to

organize into a higher level of

organization

and then that coordinates their action
in a different
way and another way to say it would be
like collective behavior and
individuals of similar or different
kinds but at the very least
interacting markov blankets that enact
organizational schemes so people who
wouldn't have talked
but then in a comment thread they're
arguing that's like
an organization that wouldn't have
existed unless there were that markov
blanket
interface unless there was that shared
niche so this doesn't have to be all
hexagonal and crystalline this could be
very chaotic and online
paper eight was really a great one
and thanks blue for suggesting it it was
called scaling
active inference as an archive paper and
alec really helped through email and
also
on the discussion itself with clarifying
several aspects of the paper
and this summary figure captures the
results so let's go through the results
and then talk about maybe how the model
did what it did and the task that these
results are based upon
is the mountain car task which is on the
top left and this mountain car
is trying uh is rewarded when it reached
this is when it reaches this flag and
it's not rewarded when it's not
and it has the affordance of either
having the engine push forward or push

in reverse
but it doesn't have enough horsepower or
whatever to
just drive up this hill so it has to
rock
back and forth many many times in order
to
ever get enough momentum to reach the
top of the hill
the issue is that because none of these
states are rewarding
as it starts exploring the local space
of policies it can enact like let's do
three seconds forward or let's do
always forward or let's do forward and
then backwards alternating for 10
seconds so then in 20 seconds
whatever state space it starts exploring
with these alternative reinforcement
learning algorithms
it turns out that that local state space
of policy
keeps it relatively in the bottom
dwelling of this basin
and that's because these reinforcement
learning algorithms kind of require
reward they need that sugar along the
way to start learning by example
and they can overlearn on initial
example successes
that's also an issue but they do need
some
help along the way to find some success
and both of these algorithms the ϵ -greedy a little bit better so
than just the
straightforward reward learner uh
they both uh managed to

not explore a full range of the
locations

because yeah they're sort of dwelling in
this like bottom zone

active inference model on the other hand
which as we've also been discussing
takes sort of a different
approach to this explore exploit
trade-off

it was able to explore a broader range
of positions

and a broader set of velocities so it
was just accelerating faster to the
bottom and then it was able to
get up further on to the other side and
there was great discussions there
about the novelty driven

versus the exploit driven and there was
also another figure that had to do with
a different set of tasks that was also
pretty informative but this model
was um helpful to go through
and it just really highlighted the value
to our group of being really specific
about what the models were doing and
also

i think a lot of those who didn't have
that much experience with the machine
learning

were able to get a peek into the state
of the art

of machine learning it's a paper on
archive and it's being
discussed by machine learning people and
see what

these algorithms were and weren't
being tested on what they did or didn't
include and so that

insight was just helpful as always
number nine was the projective
consciousness model
and phenomenal selfhood that was a fun
sequence
and the two of the authors visited for
one of the streams and it related to
the intersection of a few different
topics which were
just to pick the most key ones were
projective geometries
and cybernetics and integrating them
under free energy principle
with active inference as a process
theory so the
projective geometry idea is that there's
different geometrical spaces just like
you could have
a flat piece of paper flat plain or you
could have a sphere
or you could have a taurus or different
kinds of
geometries on different types of
surfaces you can have different
projections
amidst those different types of
geometries
some of those projections are
two-dimensional
like the one on your screen and your
brain does
uh generative state modeling to infer
depth others are
actually three-dimensional um but
there's
a few aspects of this projective
geometry
that uh equate or

uh correlate very strongly with in their
mind what were the fundamental features
of
our conscious awareness which is that it
has
a non-arbitrary origin that appears to
be somewhere in the visual
field and that objects that are far look
far and objects that are close look
close
you know who knows and uh
they tie the functional attributes to
consciousness on one half
to the projective geomet projective
geometry
uh features as they're differentiated
from euclidean features geometry
and then on the other half of the
functional features the goal seeking
components were tied to the cybernetics
and so the cybernetic or goal seeking or
recalibrating systems approach learning
single loop double loop triple loop
learning whatever it is that humans are
doing
that cybernetics approach also
constitutes some of the features
functionally of consciousness
this figure conveyed how uh if you have
a two-dimensional projection that
doesn't
have any perspective cues then
it's an undecidable inference there's
two equally good solutions
to whether this almost
hints at being a cube but can oscillate
between whether it's a cube facing
top down or which way it's facing

depending on how you see it
so one thing to consider is people's
priors on wire
cubes makes them see a cube in this
projection because it's just a
two-dimensional projection so that's the
first
thing is that we unpack these line
drawings into
contextualized especially if it says
which way do you see the cube facing
it biases you to think that you're going
to see a cube instead of like a logo for
a
company or something like that and which
still could be a cube
then once you're on the cube wavelength
it kind of oscillates
or maybe it doesn't depending on
different people uh which way it's
facing
and that's because the two-dimensional
projection is not
telling you which side you're on but
when you're in a room it doesn't seem
that way
and it feels like uh you're in a room
and you know that you're in a
rectangular room even if the angles look
off
and so they relate that to the goal
seeking
cybernetic nature and the projective
geometry and
it was a great and really educational
discussion with the authors as well
the tenth paper and the graph
that will that i referenced earlier that

we would come back
to is the variational approach to
scripts which was a great
sequence of discussions as well because
we
had this kind of a mathematical figure
but it was really
a paper that cited and drew heavily on
the social sciences
and on the social scripts concept and
shout out to mao for
coming on and for um yes you asked
would you be open to talking about a
colloquium i would just
put the details and i'll do it and
in this paper uh it started with
a review of the literature on the
different ways that
this scripts concept had been used and
it compared it along two primary axes
which is strong versus weak
and internal versus external and so
strong versus weak
was related to whether it was a very
deterministic
understanding of a script like a very
ritualistic or stereotyped performance
versus a very weak script something that
was more probabilistic
or potentially
not predestined in some way that's the
strong weak axis
then the internal external axis which
was related to the
beyond internalism and externalism paper
was talking about whether there was a
prioritization of internal factors or
external factors so for example when

talking about human behavior
and this reminds me a lot of the book of
professor longino
studying human behavior about aggression
and sexuality
and about how different schools of
thought and different kinds of
biologists different philosophers and
scientists
different people believed that we could
um
prioritize the genetic causes versus the
physiological
versus the childhood the psychological
versus the social so broader or
structural factors
and her book is just a phenomenal
look at that so that's extremely
recommended
but in this script paper
those two dimensions of script strong
weak and internal external
were integrated through this common
skeleton
scaffold so this is how the script
framework
is going to be partitioned just
qualitatively but eventually in some
places quantitatively in a way where
there'll be situations where
internalist causal chains are relevant
where externalist supporting causal
chains are relevant
like there's times where a bird just
hits the windshield that's an external
cause
that's not an internal cause you didn't
cause that on your windshield

but there's times where car injuries
might be because of something
like that happens to a neuron and then
that leads the person to change their
driving behavior and hurt their car
so there's so many different ways where
you would want to be talking about the
person driving their car
and not get caught up in this internal
external strong week
we want to have a model that is all over
the map on those two dimensions
with a common and comparable framework
and
this is just something that we return to
again and again which is that there's
time moving from left to right so time
step one two three
and at each time step there is
observations that are coming from the
world
so that's like photons hitting the
retina or things touching the
mechanoreceptors
and then the organism in this case or
whatever the system of interest is
is doing a state estimate on those
observations
and the state estimate is like a hidden
state of the world that is being
confirmed or supported or
not through a single or a multi-chain
multi-scale mapping from these
observations
so that's why once you're an adult and
you have object permanence even when you
cover
up your visual field or your blink you

um can still have a
understanding of where objects are
because you're doing an underlying
generative model with a state estimate
of the world
a is the matrix that maps state
estimates the world to observations
d are the predictions of initial
conditions and that plays into your
initial state estimate in modeling the b
matrix is a prediction of how states
in the world change through time and of
course that can be conditional on
other features the b matrix
which is uh able to be the same through
time
in a very simple model or could evolve
through time
is influenced by policy and so the
example
that comes to mind is always like if i
take the policy of exercising every day
that's going to change the probability
of my state estimate for health moving
from here to here to here
so our policies change how the states
the world
change so instead of thinking of
policies as just
connected directly to states of the
world changing you know you can't just
grab the gdp
and just yank it down but you could
enact a policy
that would result in the hidden states
changing in a certain way
and you don't need to specify every
single way that that happens down to the

protein vibrating

because your model is a coarse graining

of course it is

power policy is selected this is where

the control theory

and the cybernetic side quantitatively

comes from

and the social script side this is uh

sort of where the social pressure

versus individual preferences stories

come into play

this g is the expected free energy and

it's the expected free energy

minimization

that leads to policy selection as given

the niche and my understanding of it

what is the most likely course of action

for me as that kind of agent

to engage in and there's a lot of other

natural language ways to say it but in

the end it's about the specifics of the

model

and here's just a wireframe model but it

would always come down to the specifics

of the

model and this paper was just i remember

a fun discussion because we talked about

a lot of real world

examples of where scripts come into play

so a lot of people had interesting

perspectives there

and also about how active inference and

its understanding or its framing of

narrative and syntax and semantics

could relate to online communities or

real world

communities paper 11

which was just these last couple of days

and weeks

was sophisticated affective inference

simulating anticipatory

affective dynamics of imagining future

events

and this was actually a figure from one

of its supporting citations

but it was brought up in the 11

discussions and it shows this

nesting of models that we just discussed

on the previous

slide with the states and observations

policies affecting how states change

free energy influencing that

we see the same structure with a little

bit level one more level

of detail so here we have the initial

state

we have the state how it changes through

time with the b matrix

and the a matrix which is the mapping

between the states and the observations

there's π the policy which is being

connected and influencing how states

change through time so that's this

bottom part here

here's g and that's how it was laid out

here

now we have g but we're going to add a

few more auxiliary variables

so the first auxiliary variable here is

c which is the preference

vector so the preference vector is truly

an inextricable

piece of this whole thinking so

not like other statistical methods which

sometimes can remove

one from their preferences by making

certain types of things
ranked implicitly by certain features
like their abundance or
other aspects the preference vectors
whether stated or realized or or
idealized
are an integral part of these
equations because how policy is selected
is about preference it is also related
to
e which is the prior of affordances
for policy so that's related to these
ideas we talked about
like the um field of affordances
and the skilled intentionality framework
and the entire uh umwelt
and ecological psychological angle that
is so
nicely uh introduced by axel constant
yellow bruinberg and others this
field of affordances is like the priors
over what action is possible
so if all of a sudden some i heard some
noise
and that makes the likelihood of some
generative model like there's some type
of thing happening outside of my door
in my office changes the likelihood of
that from one in a million to
one and five and then i start thinking
about policy
well policy thinking should be reflected
by a weighting in what is possible it
shouldn't be like i should be sitting
here thinking about teleporting out of
my office
because it's not a policy i can engage
in so that's down weighted we can think

of and so

the ways that policies are selected is

through the free energy

minimization or it's expect the

minimization of its expectation

as it relates to the prior and the

field of affordances and the preferences

this free energy minimization

is mediated by a

prior and a posterior over precision and

so the prior and the posterior over

precision β and γ

um are both related to precision but it

has a prior and a posterior

component so that means that it's

getting updated through time so this

precision parameter is being updated

through time as

things change as observations come in as

actions are performed

and it relates to how confident the

organism is

in its model of cognition so that's why

this

m4 is called the generative model of

implicit metacognition

because it's a confidence estimator

which could be delusional or it could be

adaptive

or some other adjective in

the organism's uh model which is a

phenotypic congruent as in i am that

kind of agent

model of actions that has to do with

the way that hidden states of the world

are reflected in observed states

so if you believe that you are a certain

type of agent but seeing something

different in the observations
then that is going to lead to a
discongruity and an
imprecision in this total stack as it
moves through time
so cool that was the 11
papers here we are in 12.
12 which doesn't have a paper i put a
clock
because again it strikes midnight and
uh only once on 2020 and here at active
inference lab
we learn around the clock what does that
mean
that means we aim to include people
across
areas and time zones and
we provide shared structure shared
structure
is the enabling architecture for meaning
shared structure is what helps us
synchronize it really
helps us with our shared sense making
and we return to attracting sets of
ideas and other kinds of attracting sets
again and again and
i'm gonna just look into the live chat
and
somebody with the uh initials a s wrote
hi it's my first time here i've read
about carl fristen and the free energy
principle
what are the resources that i need to
get into to get involved
this is a great question and i
appreciate it
it is an attracting set that we should
always be returning to

which is how can we get involved how can
we participate
and then on the other side of the
blanket
how can we make it easier to participate
so
the answer to that question is go to
activeinference.org
and there hopefully you will find
starting resources
on how to learn more about
this area so one resource is this
live stream and the curriculum of papers
that we discuss
and then um in the upcoming year
we're going to be hopefully developing
more types of educational resources
and if you go to the 2021 podcast
uh you know live stream calendar
spreadsheet that i showed
earlier the the short link
uh you'll see on the top right two
resources that was of baron and jared
that
contain a ton of resources so another
thing that i would highlight here is
that
uh if you're really curious about this
just
stay in touch so get a keybase username
and join our team or whatever platform
we're using in the future
or email us but just if you're curious
about something
email us or put it in a comment in a
thread
and we'll develop better platforms for
curating and connecting

people on these questions but stay involved
and curious and in the community and read the articles at whatever level you feel comfortable with them and it will always be helpful to have somebody to read them with and a group to read them with whether it's on a live stream or whether it's not on a live stream whether it's just on email and um if you don't know what to read next or what video to watch next you should just ask the question like hey i'm really into this and that i know about this but i'm curious about that can anyone recommend me a paper to read or i don't know if i could read a whole scientific paper right now but can anyone recommend me an article about this or any other kind of request so i'd say that's the best way to get involved is activeinference.org will always be the most updated thing and we're going to make that a lot more informative in the future and you can read through things that we've discussed on this uh active inference live stream and then really just be socially communicative and ask people just through private email to us or to anyone or just a public comment especially if you think it's a question

that other people might have
um if that answers your question please
let me know otherwise anyone else free
to
ask questions here's two pieces of
feedback from
the community that were sent in the days
leading up to
preparing this and i'll start with the
top one
saramono wrote i just want to see a kind
of toy model with a number of relatable
variables that are tenable and tractable
one of the challenges i have with
understanding this stuff is it seems
like you find out what the variables are
through the model and then find out
there's a gazillion of them and it's
hard to trace how things converge
maybe i'm wrong or this is too naive of
a wish but i want to be able to step
through python matrices
to see how the model changes at each
stage yes good
point good question so it points to a
need for
interactive notebooks and other kinds of
technology
that allow us to very uh granularly
play with and see and explore
these models it's one thing to see it
graphically laid out in a figure as we
had in some of these previous slides
it's a whole another thing as you put it
for those variables to converge
on a functional state so that's a great
point and also the tractable and tenable
would be cool because then people could

apply it to different kinds of models
so we wouldn't want this notebook to be
full of esoteric
code it should be well commented could
it be in multiple languages multiple
computer languages or could it be in
multiple natural languages with the
commenting
could it cross render in a way that
helps people learn
could we collab on the code
we want to be able to make that
experience helpful because again
it's not just what to read it's also how
to be communicating and
being the kind of person who's coming to
understand it and
everyone's going to be learning the math
forever i mean you could get your phd
on a sub sub sub math area so
we're talking about multiple areas of
math so no one's going to be that deep
in the game
so it's about the community so that's
the cool
thing is just you'll come to understand
what you're interested in doing is it
applying it to
your area is it um just looking at what
kinds of projects people are interested
in
in for the rcps it includes youth
education
theory crafting entrepreneurship study
groups so there's a lot of things that
people are interested in doing to say
nothing of
robotics and other deliverables so

um i hope that a common uh resource
is the kind of computational clarity
that ceremony is getting at though so
thanks for that
mao wrote um that
it would be nice to see adapted
implementations and cross-disciplinary
lingua franca
which is great because it's um that's
like the french
language the french tongue yet it's an
english
communication ostensibly but it just
means common speak
um and that is part of the question
and it's fun to cross languages how do
we have
the right trade-off with accessible
but specific language as well as
non-violent
communication and very accessible and
intercultural
norms so it's not just about the
rigor and the precision in language
because you can go down that rabbit hole
and not come out just like you can go
down the other side
of caring about how the words are
strictly perceived by an infinite
set of possible external actors
somewhere in the middle
is productive research that's also going
to be
really productive for hopefully a wide
variety of people
and it's a a good piece of feedback
because one of the lingua francas if you
will

is practice and adapted implementations
so the cross-disciplinarity the
interfaces between the different
disciplines
are not just the undergraduate majors in
the university
and having the biologists the
mathematicians and the social scientists
come together
it's also about understanding theory and
practice
and understanding how people who want to
be interacting with
active inference from so many different
perspectives so just like
there's biology education of all kinds
and
could and should be more adult education
in all these different areas
active inference hopefully will be
something that's a lifelong
learning journey and i think especially
with
some of these um things that people
wrote in the rsmps
with the youth education i'm just really
excited
to understand what that could look like
too because we could all
use youth education and
um uh as wrote thanks that was helpful
i'll try following thank you
cool so now on to just the random
section something uh i can't remember
where i
heard it but it was this idea of mirrors
and windows
and two kinds of interfaces and

the window is one that you look through
and the mirror reflects us bizarrely now
uh what we what we see is a generative
model of the world
so first off how weird is it that our
generative model of the world
can either make a surface transparent or
not
i mean shouldn't you be able to imagine
what's on the other side of a wall so
maybe
that's what looking through things is
sometimes
in the visual dimension or not when
different
levels of precision in different parts
of the world
peer through so there's so much
that that could relate to but another
way that this relates is
in the discussion i think in nine with
this
transparent concept uh the transparent
concept the water we swim in
is the things that are so fundamental
that they don't seem like they're up for
debate or distinction which is the whole
nuance of zenko on and
of this consciousness studies area is
that it's
it's the very subtle and potentially
even ineffable
aspects of consciousness that are its
fundamental attributes
and potentially the most fundamental
attributes are those which are not
directly experienced at least under
normal

consciousness so there's this
distinction between the window
which is the concept is it's us
seeing things from the outside in
but um it's almost like your eye is
radiating
outwards from the generative model
distribution perspective
as opposed to the mirror which reflects
us but it reflects us potentially
bizarrely
reflected up down left right who knows
all kinds of bizarre reflections in the
echo chamber
and then that made me think about the
two common sayings
that uh if you're in a glass house don't
throw stones
and then the second one was a hall of
mirrors which is
uh being in a diluted state so the it's
funny that the
common wisdom around these two kinds of
interfaces
for the glass for the well they're both
glass
but for the window it's saying if if you
live a transparent life
you shouldn't commit aggressions
or transgressions and then on the other
hand so if your interfaces are too
transparent
you have to live increasingly
responsibly
on the other hand if you live in a hall
of mirrors
it doesn't say don't throw stones you're
just in a delusion

any stone you throw would actually
potentially even draw you out of the
illusion
by disrupting some of the feedback
mechanisms
so pretty interesting and uh
just 20 20 on the close video
just really time to look back and look
at the self
look into the future these are always
critical things but they're taking on
special relevance now
just one little
foray into the synergetic realm and this
is drawing on
some of the dissertation and work by
dr cheryl clark on the social
synergetics site
and i just happened to be reading this
and i found
this figure and i just
was already planning to do the clock
with the 12 and so i just thought
it looked like the clock and so it would
be cool to
go there and uh
i wonder if active inference can be
related to synergetics so what is
synergetics really there's so much to
say here but synergetics contains
elements
of a few different areas of mathematics
and i started with this quote unity is
plural and at minimum two
there's many kind of classic quotes of
synergetics which is developed
primarily by buckminster fuller no
longer here physically

and uh it really reminds
one that the basis of this way of
thinking is relational
so that relational stance which is
shared with active inference at a very
kernel level
is going to be implemented in a couple
of different ways
so in synergetics the key pillars are
related to
the areas that we would consider
geometry topology
or graph theory or network science it's
also related to
vectors and dynamic systems as well as
structural modeling
which was developed by buckminster
fuller through the engineering practice
of
tensegrity tensegrity modeling but
there's a few other
interesting and important pieces of
synergetics
and one of them is that the tetrahedra
is the minimal system
not the cube and the tetrahedral
uh focus pervades several different
domains one of them is that there's a
focus
on the 60 degree angle as being primary
rather than the 90 degree
angle so the tetrahedra is the minimal
system and this plays out in a million
different ways but the tetrahedra on the
screen
relates to plato's triad and
it shows on the right side this
triad of the true uh the good and the

beautiful truth beauty and symmetry any
three you want relate them however you
like and then
the tetrahedral edition of the self so
this is the perspective this is the
relational component and
quite interesting to think about some of
the grand transformations that are
occurring
at this time in world history from
the perspective of synergetics and the
introduction of the self
into this triad and
other tetrahedra can be imagined uh
crystals
buildings social relationships
what really makes me curious is that in
active inference on the coarsest grain
there
are also four states that's that four
state model
and those are the two kinds of external
and internal states
and then the uh sense in action states
so there's a unity in those x internal
and external states that's unity is
plural
at minimum two internal and external and
then
there's the interface and then the next
level out is four not
three the interface is not one part the
interface is two parts
so you go from unity as plural and at
minimum two internal and external states
to the triangle which is stable but as a
modeling concept but
doesn't have space enclosure it's not a

minimal system

the minimal real specified system is the

tetrahedron is the four state active

inference

model so from a totally different

perspective

non-geometric perspective but

intriguingly using information geometry

fvp free energy principle has converged

to somewhat of a synergetic perspective

on how unity is plural and how the

minimal system is potentially a purely

relational tetrahedra

so that is quite interesting big if true

also from her dissertation this figure

of a tetrahedra

relates how qualitative concepts can be

captured

within synergetics so here

the edges of the tetrahedra or the

vertices

are vision reflected by a mission

statement

tools hopefully the mission statement is

something that we will write together as

a

lab and as a community with feedback as

well

on the right side tools and resources

that's one of our projects

and hopefully the tools that we build

will be for ourselves and for everybody

and the resources that we make

will be for everyone so that's education

as well everything is tools and

resources

on the left side design in the action

plan to be

bringing people together successfully
we're going to be needing to co-create
co-design and then enact our action plan
shared joint action it doesn't mean that
everyone's going to do the same thing
but we are going to co-create and enact
together
and then on the bottom evaluation and
supervision
and this kind of mentorship peer
mentorship
whatever it happens to be considered as
or called
facilitation is how we're going to
make sure that this whole tetrahedra is
coherent
another feature of synergetics and
tensegrity is this
focus on both tension and compression
and so the idea that bucky had
was that the uh buildings on earth they
require gravity to pull them down like
the building i'm in if it was on its
side it would just crumple
from its own mass and so they require
gravity in order to hold them down
whereas if you're going to be shooting
something to a different planet or
landing it from a high altitude you need
to be able to be
kind of self-supporting from any
gravitational influences
and so these kinds of tensegrity
structures
are um with both compressive and tensile
elements so like ropes
that are connected to rods and if you
just look up tensegrity sculpture or

tensegrity robot

you'll see a ton of cool robots and

other stuff

and um the big question is could we

connect it

with active inference and the fep and

here are

some people amy edmondson kirby earner

kurt mcnamara the 10 second wiki

snec the cj apparently all the groups

that he does

everyone else i don't know all of them

these are just some that i

know or have learned a ton from

but there's so much that could be

explored here

the question that we return to just so

many times

in this year and we'll return to again

is what does free energy principle and

active inference say about the

relationship between

an agent and the world and

that has come to be uh understood

in these discussions as the relationship

between

the different parts of a multi-scale

system between the psycho

psychology of being in social settings

uh

online teams computational things that

are communicating with each other

all these different ways that we talked

about how the agent and the world

interact in all the different systems

that we would want to

understand what this means in i mean

it's a general crush question and

let's continue to think about these big
attractor set
questions that will help us just always
come back to the basics and always focus
on making it the most applicable
most comprehensible and just to combine
that clock
in the clock striking midnight with
active 12.0
and the kairos so the uh
greek language has these two words for
time
related to time kronos being the
chronometer
being the time that's broken down into
quantifiable pieces
uh even taken to the extreme of metric
time
by some you know 10 hours per day 10
minutes per hour 10 seconds per minute
and then in contrast there's this kairos
and these two um frameworks for time are
so
related but they're related in this way
that
syntax and semantics are kind of that
scaffolding
and meaning are and it's not a formal
mapping as always these are just hastily
prepared
discussions just some thoughts but um
as we build active inference in the
computational
direction and implementational with
internet structures
uh and also build in the accessibility
dimension
and the communication and the education

and cross-disciplinary thinking
as brought up by the great community
feedback and so many
other people we we never um
want to stray away from the meaning
discussions
and that is uh something we'll learn
along the way how it plays out but i'm
just curious
how that will play out so i can suffice
to say
that and
wow i mean i don't have anything
prepared
from a writing perspective not going to
add any more
any notes just 20 20 was a big year
and a lot remains to be said
and i hope that everybody can
have a safe and healthy life
and i just want to keep this about
active inference so i really
just don't
want to uh predict or
review it just feels like a time that
we're all moving through in a really
unique way so i just hope everyone's
well and just
may the gradient descent be with us all
and one
also last know on 2020 is um
during this year my colleague martin
passed away
he's here and um
he was an incredible person
and he co-founded this group alias that
i'm a part of
which is a research group on

consciousness and he
was a co-editor on this book
metacognitive diversity
he was a french scholar did a lot of
field work in
the amazon and other areas and
um in 2018 in the alias bulletin
we uh martin and i interviewed carl
fristen
about free energy principle and uh
so martin was really even though his
background was in
humanities and
anthropology he learned so much about
neuroscience
and multi-scale biology and the free
energy principle
and just made it come alive and was such
a cognizant thinker so
all right thanks for participating
everyone
it was a uh good stream yeah
r.i.p martin sorry everyone but it was a
loss so
uh yeah just
intense stuff everybody lost people you
know so
anyways good times everyone thanks for
being along for the ride and uh i i
think that
we'll have a lot to do and work on
in 2021 i uh
hope we can build from what's happening
now
and build towards something that's just
better i know there's a better way
and it's going to be really important
that we get there and find it

together so

thanks again just stay in touch

get involved r.i.p martin

long live active inference peace

everyone